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EXTENDED ABSTRACT

Self-Supervised Spatiotemporal Clustering for ENSO Characterization

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1. Introduction and objectives

Identifying meaningful climate regimes is essential for understanding climate variability and improving predictive modeling, yet remains challenging due to noise and nonlinear dependencies in raw climate variables. In this work, we apply Masked Siamese Networks (MSN) to discretize daily minimum and maximum temperature fields into a finite set of automatically clustered climate regimes.

Rather than modeling exact temperature values, our approach learns categorical representations that capture dominant spatiotemporal patterns in surface temperature fields. We show that these learned regimes are physically interpretable and exhibit robust statistical associations with the El Niño–Southern Oscillation (ENSO).

By integrating self-supervised Vision Transformers with climate regime analysis, this work contributes an interpretable, data-driven framework for discovering climate states that capture both seasonal structure and long-range teleconnections.

2. Methods and data

Climate fields are discretized using a Masked Siamese Network (MSN) [Assran et al.(2022)] with a Vision Transformer encoder. MSN learns representations by matching masked and unmasked views of the same input using an anchor–target encoder framework (Fig. 1).

The learned representations are discretized via K trainable prototypes, yielding soft cluster assignments during training. At inference, the target encoder maps each input x_t to a discrete climate regime by nearest-prototype assignment, transforming the continuous temperature fields into a categorical sequence suitable for more straightforward analysis of climate regime frequency, transition, and teleconnection influences.

2.1. Dataset and Region

We use the Brazilian Daily Weather Gridded Data (BR-DWGD) [Xavier et al.(2022)] at 0.1° resolution, extracting daily minimum and maximum temperature over the Brazilian Cerrado (lat $[-22.0, -7.0]$, lon $[-57.5, -43.0]$). Each day is represented as a tensor $x_t \in \mathbb{R}^{H' \times W' \times 2}$. The temporal coverage spans January 1961 to March 2024. Years 1981 and 2000 are reserved for testing; all others are used for training.

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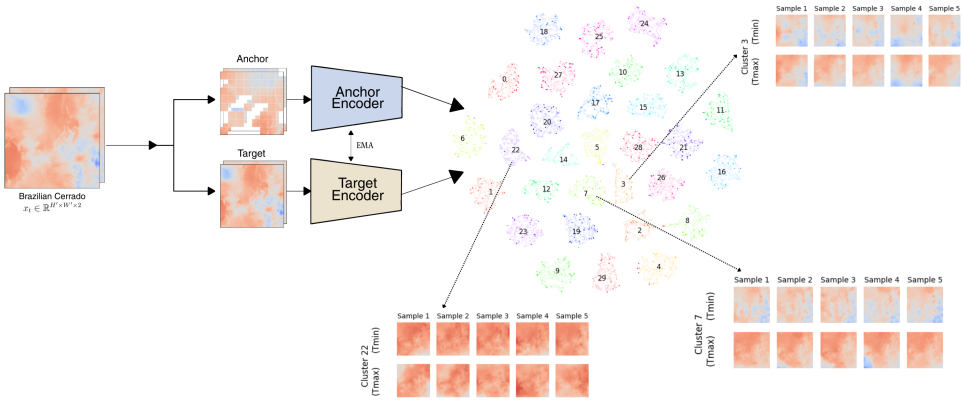

Figure 1. Self-supervised discretization of climate fields over the Brazilian Cerrado. *The input daily tensor $x_t \in \mathbb{R}^{H' \times W' \times 2}$ (containing T_{min} and T_{max} channels) is processed through a Masked Siamese Network (MSN). The Anchor path receives a masked, cropped view of the input, while the Target path receives the full view, with its encoder updated via Exponential Moving Average. Learned representations are discretized into $K = 30$ prototypes, visualized using t-SNE. Representative samples from selected clusters illustrate how the model groups distinct spatial temperature configurations into discrete climate regimes.*

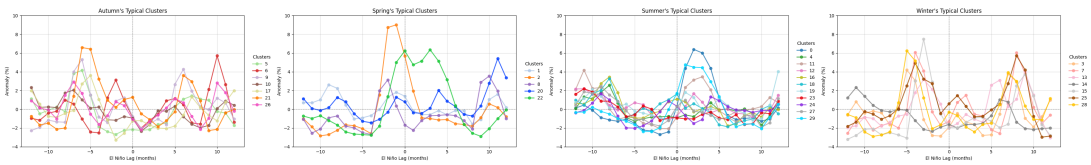

Figure 2. *Lagged probability anomaly $\Delta P_k = P(k \mid \textit{El Niño}) - P(k \mid \textit{Neutral})$ of learned climate regimes relative to the ONI for lags $\tau \in [-12, +12]$ months. Distinct regimes exhibit synchronous, leading, and lagging responses to ENSO, revealing both concurrent atmospheric signatures and precursor states that precede peak oceanic anomalies.*

3. Preliminary results and/or expected outcomes

Initial tests using $K = 30$ prototypes demonstrate that the framework successfully identifies physically meaningful climate regimes and captures large-scale teleconnections without explicit supervision.

Lagged probability anomaly analysis relative to the Oceanic Niño Index (ONI) indicates that different regimes exhibit distinct temporal relationships with ENSO (Fig. 2), including synchronous responses as well as leading and lagging behavior. In particular, several regimes show enhanced probability prior to ONI peaks, suggesting the presence of precursor atmospheric states encoded in the learned representations.

These preliminary results validate the model's potential for forecasting and establish a foundation for further investigation into regime transition dynamics and long-range teleconnection patterns.

References

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